

Flammability Testing Requirements

U.S. – CPSC 16 CFR Part 1610

- Applies to general apparel sold in the U.S.
- **Requires flammability testing unless exempted**
- Standard classifies textiles by burn speed and surface characteristics

 **Part 1610 - Specific Exemptions**

❖ Fabrics that meet a specific exemption do not require testing.

- Determined by fabric type and specifications
 - Plain-surface textile fabric or raised-fiber surface
 - Fabric weight
 - Fiber content



❖ Plain-surface fabrics $\geq 88.2 \text{ g/m}^2$ (2.6 oz/yd²), regardless of fiber content

❖ Plain- and raised-fiber surface fabrics made of certain fibers:

- Acrylic, modacrylic, nylon, olefin, polyester, wool, or any combination of these fibers, regardless of weight

U.S. Consumer Product Safety Commission

U.S. – CPSC Exemptions from Testing

- Plain-surface fabrics $\geq 88.2 \text{ g/m}^2$ (2.6 oz/yd²), any fiber
Fabrics made entirely of acrylic, modacrylic, nylon, olefin, polyester, wool
- Includes recycled polyester and recycled nylon
- Exempt if weight and surface criteria are met

EU – GPSR - Flammability Requirements

- EN 1103:2005 is widely required for adult apparel
- Recycled polyester and nylon treated same as virgin fibers
- General Product Safety Regulation (GPSR)

- **USA (CPSC 1610):**

- Flammability testing required
- Exemptions for specific fiber types & weights
- Includes recycled polyester/nylon

- **EU (EN 1103):**

- Flammability testing required
- EN 1103:2005 is widely required for flammability testing for adult apparel
- Children's nightwear & baby clothing (varies by country)
- Personal protective equipment (PPE) – EN ISO 11612
- Home textiles (e.g., curtains, upholstery – especially in the UK)

Switzerland is part of the European Free Trade Association (EFTA), **not the EU**, but it often aligns with EU standards.

EN 1103 is a harmonized European standard for flammability of textiles, and many brands use it for broader European compliance.

I believe, we require such testing to **stricter benchmarks, to ensure we comply with EU alignment**.

EN 1103 is the European Flammability Testing Standard with partial testing method of textiles **for apparel**.

EN ISO 6941 is the Testing Method referenced within EN 1103.

EN ISO 6941 is an international (ISO) standard and describes the technical execution of the flame spread test for certain range of textile materials.

Summary Table FYR:

Both of them are correlated, I think that's WHY we put both on our manual – which make sense to me.

Standard	Applies To	Covers	Remarks
EN 1103	Apparel (Europe)	Burning behavior, procedure, pre- cleaning, result judgment	Uses EN ISO 6941 for test method; product specific
EN ISO 6941	All textiles, globally	Test method for vertical flame spread	Used by EN 1103 and other standards

Flammability Testing - Mandatory Requirements Overview

Jurisdiction	Citation	Scope	Remarks
USA	16 CFR 1610	Wearing Apparel	*Class 1 textiles exhibit normal flammability and are acceptable for use in clothing. *Class 2 fabrics, applicable only to raised-fiber surface textiles, are considered to be of intermediate flammability, but may be used for clothing. *Class 3 textiles exhibit rapid and intense burning, are dangerously flammable and shall not be used for clothing.
CANADA	SOR/2016-194	Wearing Apparel	
MEXICO	No requirement	Wearing Apparel	N/A
U.K	Sleepwear	Wearing Apparel	
E.U	EU (EN 1103) Netherlands - General Product Safety Directive 2001/95/EC Norway - Directive on prohibition of highly flammable textiles Sweden - KOVFS 1985: 8 Switzerland - Verordnung RS 817.023.41; Section 5; Article 16 et seq	Wearing Apparel	See excerpt from manual
CHINA	GB 31701 – for Children and Infants	Wearing Apparel	Ordinary clothing in China does not require mandatory flammability testing.